Fruition Technologies Geese Detection

Analysis of 4 AI Detection Methods, Performance Comparison Matrix, and Implementation Roadmap (Feb 16 – May 1)

Landscape of Detection Methodologies

SINGLE-STAGE / REAL-TIME (Speed Focus)

YOLOv11-Nano
Ultra-fast, lightweight model designed for real-time edge applications, balancing speed and accuracy.

SSD (Single Shot)
Predicts bounding boxes and class scores directly from feature maps in a single forward pass.

EfficientDet-D0
Scalable architecture that optimizes both accuracy and efficiency across diverse resource constraints.

TWO-STAGE / TRANSFORMER (Accuracy Focus)

Faster R-CNN
Classic two-stage detector using a Region Proposal Network (RPN) for high accuracy, but slower.

RF-DETR (small)
Transformer-based detector utilizing attention mechanisms to capture long-range dependencies.

ViT-Det
Pure Vision Transformer architecture for detection, moving away from traditional CNNs.

SPECIALIZED / SEGMENTATION (Specific Use-Cases)

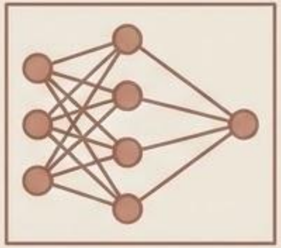
Frame Differencing
Basic motion detection by comparing pixel changes between consecutive video frames.

Hybrid (motion + bedrock)
Combines motion cues with a stable background model for robust detection in dynamic scenes.

Mask R-CNN
Extends detection to provide pixel-level instance segmentation alongside bounding boxes.

CenterNet
Keypoint-based, anchor-free approach that detects objects as single center points.

Method 1: RF-DETR (Real-time Detection Transformer)



The Concept:
Global Image Context

Utilizes a modern 'Transformer' architecture to process the entire image simultaneously, enabling a comprehensive understanding of global context, unlike older models that analyze small sections in isolation.

How it Works:
Attention Mechanisms

Leverages 'attention mechanisms' to discern relationships between pixels. It excels in cluttered environn-ments, correctly identifying a "whole bird" even when only parts, like the neck, are visible.

Key Benefits:
Reduced Latency

Eliminates the need for complex post-processing steps, significantly reducing overall system latency and making it ideal for real-time tracking applications.

RF-DETR Implementation Roadmap

PHASE 1
(Feb 16–28)

Environment Setup: PyTorch & Hugging Face library configuration.

PHASE 2
(March)

Training on Waterfowl Datasets: Utilizing COCO dataset supplemented with custom goose imagery.

PHASE 3
(April)

Sub-30ms Latency Optimization: Focused optimization for real-time tracking.

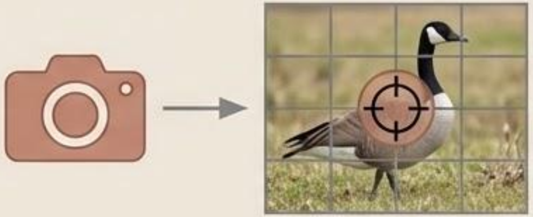
PHASE 4
(May 1)

Final Delivery: Deployment-ready RF-DETR model.

Method 2: YOLOv11-Nano (Edge AI)

The Concept: Ultra-Lightweight Edge AI

An ultra-lightweight version of the ‘You Only Look Once’ architecture, specifically designed for small computers.



How it Works: Grid-Based ‘Target Lock’

Divides the camera feed into a grid. If a goose’s center falls into a cell, that cell is responsible for the ‘target lock’. It is the fastest ‘single-shot’ detector available.

Key Benefits: Immediate & Low-Power

Designed for low-power hardware, providing immediate ‘target lock’ capabilities, ideal for resource-constrained environments.

YOLOv11-Nano Implementation Roadmap

PHASE 1
(Late Feb: Feb 16–28)

Edge Hardware Setup: Install the Ultralytics framework on a local Raspberry Pi or Jetson Nano.

PHASE 2
(March)

Model Fine-Tuning: Collect 500+ ‘Action Shots’ of geese to fine-tune the model for motion.

PHASE 3
(April)

Laser Hardware Integration: Integration with the laser signal to ensure sub-10ms response times.

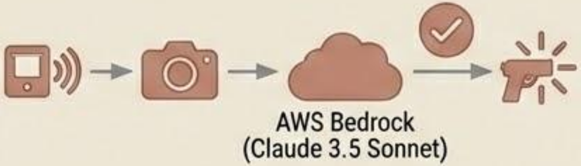
PHASE 4
(May)

Performance Validation: Final system testing and performance validation.

Method 3: Hybrid (Motion + AWS Bedrock)

The Concept: Two-Tier System

A two-tier system where a local sensor ‘wakes up’ a powerful cloud brain for confirmation.



How it Works: ‘Wake Up’ & Confirm

Simple local code detects movement (‘wakes up’). It sends a high-res photo to AWS Bedrock (Claude 3.5 Sonnet) to confirm goose presence before firing.

Focus on Accuracy & Safety

Ensures high accuracy by using advanced cloud AI for identification, prioritizing safety by confirming the target before any action is taken.

Hybrid Approach Implementation Roadmap

PHASE 1
(Feb 16–28)

AWS Setup: Setup AWS IAM roles and Bedrock API access.

PHASE 2
(March)

Trigger Logic: Develop the 'Trigger' script that only uploads frames when motion >10% is detected.

PHASE 3
(April)

Prompt Optimization: Optimize prompt engineering to ensure the AI returns a simple 'YES/NO' to reduce costs.

PHASE 4
(May)

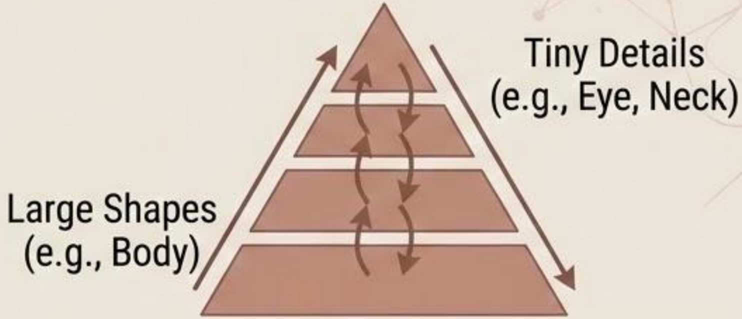
Final Report: Cost-benefit final report.

Super easy to setup and manage but may be slow tbh and also needs network connection to work

Method 4: EfficientDet-D0

The Concept: Google's Efficient Efficient & Precise Model

A model developed by Google that scales 'depth and width' for efficient mobile performance.
It uses a Bi-directional Feature Pyramid Network (BiFPN) to balance speed and high precision.



How it Works: BiFPN Diagram & Example

Simultaneously sees tiny details and large shapes, enabling high-precision detection on small targets like a goose's head or neck.

EfficientDet-D0 Implementation Roadmap

PHASE 1
(Feb 16–28)

TensorFlow Lite Setup: Setup TensorFlow Lite environment for mobile-ready deployment.

PHASE 2
(March)

Model Quantization: Quantize the model (compressing it) to make it run faster on the device.

PHASE 3
(April)

Accuracy Testing: Field testing for accuracy at different distances (10m–50m).

PHASE 4
(May)

Documentation Delivery: Final documentation delivery.

Rank	Method	Speed	Cost	Quality	Ease of Setup	**Total Score**
1	**YOLOv11-Nano**	5	5	4	4	**18**
2	**RF-DETR (Small)**	4	5	5	3	**17**
3	**Hybrid (Motion + Bedrock)**	3	3	5	5	**16**
4	**EfficientDet-D0**	4	5	4	3	**16**
5	**SSD (Single Shot)**	4	5	3	4	**16**